Instructions (read these carefully!):

- This assignment must be submitted on Gradescope by the due date. Gradescope entry code: PRXD46
- Please **include a title page** with your name and the names of your collaborators.
- Homework is graded anonymously, so please avoid putting your name on each page of the assignment.
- If you are stuck, please ask for help (from me, Sarah, a classmate, ...). Use Ed discussions! AI use is generally **not** recommended for completing the homework.
- You may freely use any fact proved in class (mention it in the appropriate spot). You *may not* freely use facts from the book or other sources.
- As a general rule, try to restrict your solution to each problem to a single page. Often solutions can be short, while still being complete. If your solution is very long, you should think more about how you might express it more concisely.

HW5: Cauchy integral formula

Due: Friday, Oct 23 by 5pm

Compute with the Cauchy integral formula.

1. $$\int_{|z|=2} \frac{z^2+1}{z-1}\,dz.$$

2. $$\int_{|z|=2} \frac{1}{(z-1)(z+3)}\,dz.$$

3. $$\int_{|z|=1} \frac{z+2}{z-3}\,dz.$$

4. $$\int_{|z|=2} \frac{z^3}{(z-1)^2}\,dz.$$

5. $$\int_{|z|=3} \frac{e^z}{(z-1)^3}\,dz.$$